PLANETARY DATA, LOGARITHMS AND MODEL TESTING ON ONE PAGE

Bode's rule: turn the curve into a line

An exponential pattern in orbital spacing becomes visible after a shift and a logarithm.

1 / BEGIN WITH THE OBSERVATIONS

Measure orbit size, then choose the index

The semimajor axis a measures orbit size. Write $d = a/(1\,\mathrm{AU})$, so Earth has $d = 1$. The rounded distances below come from the supplied data.

body	m	d	$d - 0.4$
Mercury	–	0.39	−0.01
Venus	0	0.72	0.32
Earth	1	1.00	0.60
Mars	2	1.52	1.12
Ceres†	3	2.77	2.37
Jupiter	4	5.20	4.80
Saturn	5	9.55	9.15
Uranus	6	19.22	18.82
Neptune	7	30.11	29.71
Pluto†	8	39.54	39.14

† Ceres and Pluto are dwarf planets.

Number Venus $m = 0$, Earth $m = 1$, and continue outwards. Include Ceres at $m = 3$ in the asteroid-belt slot.

The gaps grow roughly geometrically. Try an offset exponential:

$$d = A + BQ^m.$$

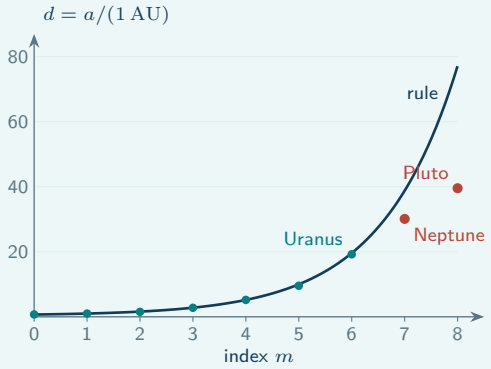
Test the *historical* offset $A = 0.4$ by plotting $d - 0.4$ on a logarithmic scale. The offset is an assumption; the straight-line fit estimates B and Q.

Mercury is a separate $d \approx 0.4$ case. Its observed $d - 0.4 < 0$ cannot enter the logarithmic plot.

2 / ORDINARY LINEAR AXES

The orbital gaps widen

Plot d against m. Points are observations; the curve is the historical Bode rule.

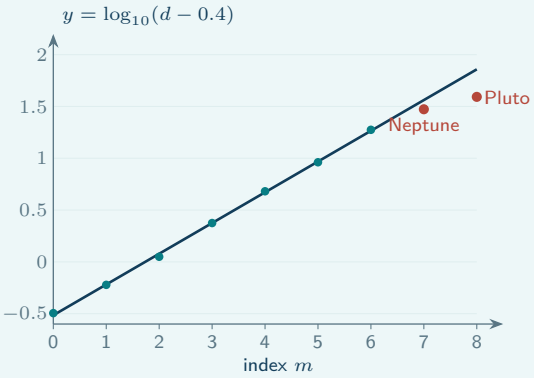


The inner bodies crowd near zero on this scale. Equal index steps correspond to increasingly large distance steps.

3 / LOG–LINEAR COORDINATES

Subtract the offset before taking logs

Plot $y = \log_{10}(d - 0.4)$ against m. Equal steps in y represent equal ratios in the shifted distance.



The line fits Venus through Uranus ($m = 0$ to 6) and is extrapolated outwards. Neptune and Pluto fall below it.

4 / READ THE SLOPE AND INTERCEPT

Recover the exponential from the line

With the offset fixed at 0.4, least squares gives

$$y = sm + c, \qquad s \approx 0.29668, \quad c \approx -0.51510.$$

Since $y = \log_{10}(d - 0.4)$, undo the logarithm:

$$d = 0.4 + 10^c(10^s)^m$$
$$\approx 0.4 + 0.3054(1.9801)^m.$$

The coefficients are close to the simple historical rule:

$$d_m \approx 0.4 + 0.3 \times 2^m$$

For that rule, the exact log-line has slope $\log_{10} 2 \approx 0.30103$ and intercept $\log_{10} 0.3 \approx -0.52288$.

5 / KEEP THE AWKWARD OBSERVATIONS

A useful pattern has limits

Keep the same index sequence when testing the historical rule:

body	m	observed d	rule d
Uranus	6	19.22	19.6
Neptune	7	30.11	38.8
Pluto	8	39.54	77.2

Skipping Neptune would move Pluto to $m = 7$, making its apparent fit much better. That changes the test.

Bode's rule is an **empirical spacing pattern**. Its offset and indexing are choices in the model. A nearly straight graph does not establish a physical cause or guarantee the next orbit.

Data. `Bodes&MilankowitzCycles Law.xlsx`, `student_Bode`, rows 4–13; observed distances and finite indices retained. The fitted line uses only $m = 0, \ldots, 6$, with equal weights in log space.

Further reading. W. Hayes and S. Tremaine (1997), *Fitting random stable solar systems to Titius–Bode laws*, arXiv:astro-ph/9710116.

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